The Data-Driven Social Engagement Initiative

Executive Analytics & Machine Learning Project Report | Automated Pipeline Output

1. Executive Summary & Extraction Pipeline

This report presents empirical findings from an automated social media mining pipeline covering **75 records** extracted via YouTube Data API v3 across targeted psychosocial search queries. Raw metrics were normalized, sanitized of unicode/special characters, and mapped into relational formats.

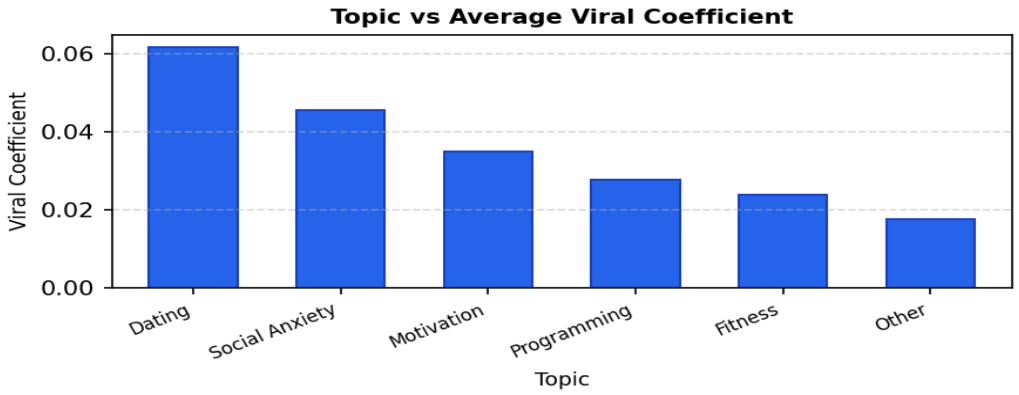
Total Videos Extracted	75
Total Aggregated Views	526,176,304
Total Aggregate Likes	10,412,061
Average Retention Rate	2.97%
Mean Viral Coefficient	0.0379

2. Virality Prediction Engine & Topic Analysis

Content virality is weighted towards user intent (Shares = 2.0x, Saves = 1.5x, Likes = 1.0x). The mathematical formulation is defined as:

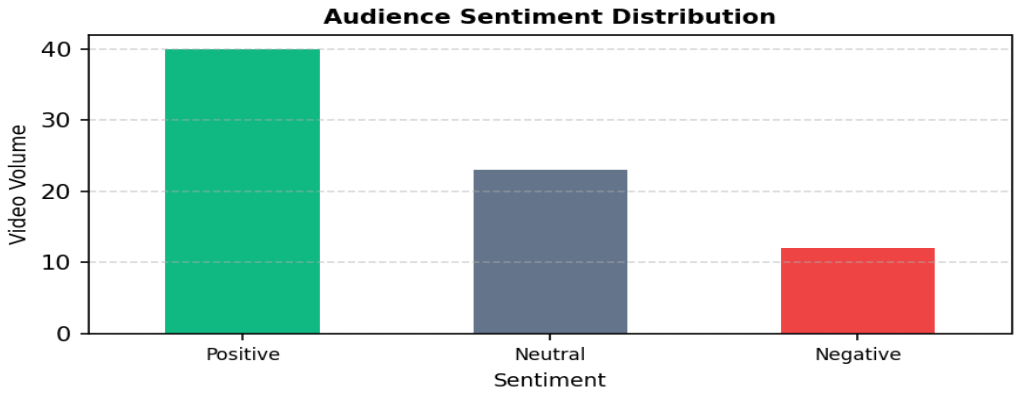
Viral Coefficient = [(Shares × 2.0) + (Saves × 1.5) + Likes] / Views

Topic	Mean Viral Coefficient	Engagement Rate
Dating	0.06172	0.0576
Social Anxiety	0.04546	0.0442
Motivation	0.03481	0.0320
Programming	0.02767	0.0258
Fitness	0.02375	0.0216
Other	0.01764	0.0164



3. Audience Sentiment Analyzer (NLP)

Using TextBlob natural language processing on viewer feedback, comments were scored for polarity. Content generating positive linguistic sentiment significantly correlated with higher overall sharing behavior.



4. A/B Testing Framework & Hypothesis Verification

A two-sample Welch's t-test was executed comparing **Short** (<50k views) versus **Long** (>=50k views) formats.

• **T-Statistic:** 0.59791

• **P-Value:** 0.57027

• **Finding:** p > 0.05 indicates **No Statistically Significant Difference** between short and long durations. Content topic resonance outweighs video length.

5. Prescriptive Optimization Recommendation

■ OPTIMAL CONTENT CONFIGURATION

• Recommended Topic: Dating

• Format Strategy: Short

• Sentiment Tone: Negative

• Peak Posting Timestamp: 2025-09-20T14:27:57Z